

Mid-Semester Makeup Test

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Course No./Title : EEE F311/Communication Systems DATE: 19th Oct. 2016

Max. marks: 50 Test duration: 90 Mins.

Important instructions:

- Exam is fully closed book.
- Write correct tutorial section number as well as instructor's name on your answer book.
- You can answer questions in Part-III in any order. But, provide answers to all subparts within a question at one place.
- Each question is marked with the number of marks assigned to that question.
- Give all your answers with proper units.
- In all sketches, include x-label, y-label, figure caption. Incomplete figures will not get full credit.

Part-I: Multiple choice questions (8 marks)

Note: Each question carries ONE mark. Write the right answer in your answer book.

1. A frequency modulated broadcast carrier is deviated by ± 37.5 KHz. Furthermore, the frequency of sinusoidal modulating signal is 15 KHz. The modulation index is equal to

A. 5 B. 2.5 C. 25 D. 50

2. Which of the following continuous-wave modulation schemes is used in commercial analog video broadcasting ?

A. Conventional AM B. DSB-SC C. VSB D. SSB-SC

3. In superheterodyne receiver, which is known as the front-end of the receiver ?

A. IF section B. RF section C. Mixer section D. Audio amplifier section

4. The Hilbert transform of $m(t) = \frac{2}{t^2+1}$ is given by

A. $\hat{m}(t) = \frac{1}{t+1}$. B. $\hat{m}(t) = \frac{2}{t+1}$. C. $\hat{m}(t) = \frac{2t}{t^2+1}$. D. undefined.

5. Let W is a standard normal random variable. What is $\mathcal{P}(|W| < k)$, where k is a positive real number ?

- A. $1 - 2Q(k)$ B. $2Q(k)$ C. $1 - Q(k)$ D. $2Q(k)$

6. The random process $X(t) = \cos(\omega_c t + \Theta)$, where Θ is uniformly distributed random variable defined over $[0, 2\pi]$ is a

- A. Widesense stationary random process. B. Stationary random process in strict sense.
C. Ergodic random process. D. All of the above.

7. The autocorrelation function of the random process in Q. (6) above is equal to

- A. $0.5 \cos(\omega_c \tau)$. B. $0.5 \cos^2(\omega_c \tau)$. C. $0.5 \sin(\omega_c \tau)$ D. $0.5 \sin^2(\omega_c \tau)$.

8. The mean square value (MSV) of the random process in Q. (6) above is equal to

- A. 1. B. 0.5. C. 0 D. 2.

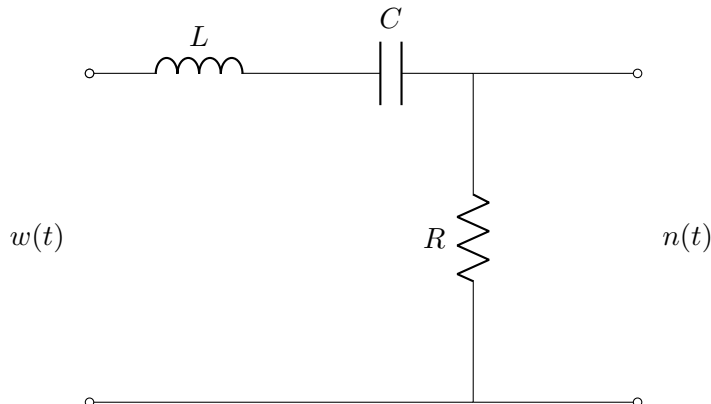
Part-II: True or False (2 marks)

Note: Each question carries $\frac{1}{2}$ point. Just indicate 'T/F' ('T' for True statement; 'F' for false statement.).

- 1) Envelope detector is a non-coherent detector.
- 2) For distortionless transmission of signals through an LTI system, the amplitude response must be flat and phase response should be non-linear.
- 3) All random processes that are stationary in the strict sense are widesense stationary.
- 4) In frequency division multiplexing, band-limited baseband signals occupy disjoint frequency bands.

Part III: Solve the following.

Q. 1. Consider transmission of white noise through a high quality tuned filter shown. Input to the filter is the white noise which is denoted by $w(t)$.



Answer the following questions.

- (a). Determine the transfer function of the filter $H(f)$. [2 marks]
- (b). Define the cutoff frequency $f_c = \frac{1}{2\pi\sqrt{LC}}$ and the quality factor $Q = \frac{1}{R}\sqrt{\frac{L}{C}}$. Express $H(f)$ in terms of Q and f_c . [2 marks]
- (c). What is the value of input spectral density ? Sketch it. [1 mark]
- (d). Express the output spectral density and sketch it for $f > 0$ and $f < 0$. [3 marks]
- (e). What are the mean value and maximum value of the output spectral density ? [1 mark]
- (f). Sketch spectral density of in-phase and quadrature components of $n(t)$. [1 mark]

Q. 2. A 10 MHz carrier is frequency modulated using a message signal $m(t) = A_m \cos(10^3 \pi t)$ volt. The resultant FM signal has frequency deviation of 5 KHz.

Answer the following the questions.

- a). Compute the modulation index of the FM wave. [1 mark]
- b). The capture range is the frequency range for which phase locked loop (PLL) will lock from an initially unlocked state. Capture range of PLL used in the FM system is equal to twice the geometric mean of modulating signal frequency and frequency deviation. Compute the capture range in Hz. [1 mark]
- c). Derive the expression for FM signal in standard form. [2 marks]
- d). Compute approximate bandwidth of the FM signal. [1 mark]
- e). If the transmit power of the FM signal is 20 dB, compute peak carrier amplitude. [1 mark]
- f). Express the wideband FM signal in terms of Bessel function. [1 mark]
- g). If the frequency sensitivity is 2 KHz/volt, compute the peak amplitude of the message signal. [1 mark]
- h). Draw the magnitude spectrum of the wideband FM signal. Indicate amplitudes of frequency components. [2 marks]

Q. 3. Spectral density of a noise random process $n_i(t)$ is given by

$$S_{N_i}(f) = \begin{cases} 10^{-16}, & |f| < 10^8 \text{ Hz}, \\ 0, & \text{otherwise} \end{cases}.$$

Answer the following questions.

a). Sketch $S_{N_i}(f)$. [2 Marks]

The noise random process is applied to an ideal flat bandpass filter with center frequency 50 KHz and bandwidth 2 MHz.

b). Sketch the ideal bandpass filter frequency response $H(f)$. [1 mark]

c). Sketch the spectral density of the bandpass filter output. [2 marks]

d). What is the mean square value of the output noise process $n_o(t)$? [2 marks]

e). Sketch the in-phase and quadrature components of output noise process. [2 marks]

f). Determine the impulse response of the filter. [1 mark]

Q. 4. Consider a random variable Y defined by the double exponential distribution:

$$p_Y(y) = \frac{1}{a} \exp\left(-\frac{1}{b} |y|\right), -\infty < y < \infty,$$

where a and b are positive constants.

Answer the following questions.

a). Determine the relationship between a and b so that $p_Y(y)$ is a probability density function (pdf).

Sketch the pdf. [3 marks]

b). Determine the probability that the random variable Y lies between 1 and 2. [1 mark]

c). Compute the mean value and mean square value of Y . [3 marks]

d). Find expressions for the cumulative distribution function (CDF) and complementary CDF of Y . [2 marks]

e). Find moment generating function (MGF) of Y . [1 mark]

□ **END OF QUESTION PAPER** □

Answers

Part I

1. B 2. C 3. B 4. C 5. A 6. D 7. A 8. B

Part II

1. T 2. F 3. T 4. T

Part III

Q. 1. Ans. (a). The transfer function $H(f) = \frac{R}{R + j2\pi fL + \frac{1}{j2\pi fC}}$.

(b). The transfer function in terms of quality factor Q and cutoff frequency f_c is given by

$$H(f) = \frac{1}{1 + jQ \left(\frac{f}{f_c} - \frac{f_c}{f} \right)}.$$

(c). Input power spectral density $\mathcal{S}_w(f) = \frac{N_0}{2}$. This is plotted in Figure 1.

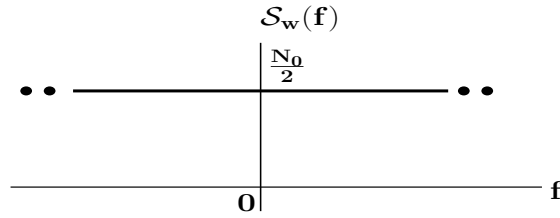


Fig. 1: Spectral density of white noise.

Output spectral density $\mathcal{S}_n(f)$ is given by

$$\begin{aligned} \mathcal{S}_n(f) &= \frac{N_0}{2} |H(f)|^2, \\ &= \frac{\frac{N_0}{2}}{1 + Q^2 \left(\frac{f}{f_c} - \frac{f_c}{f} \right)^2}. \end{aligned}$$

The sketch of output spectral density is shown in Figure 2.

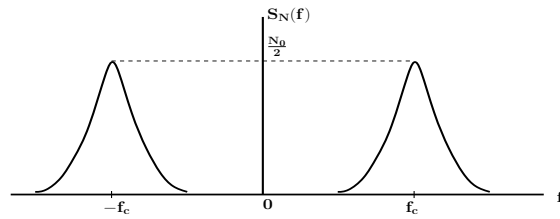


Fig. 2: Spectral density of the output of RLC filter.

e). The mean value of output spectral density = 0. The maximum value of output spectral density is $\frac{N_0}{2}$.

f). By symmetry of $S_N(f)$, spectral densities of in-phase and quadrature components of output noise are equal as shown in Figure 3.

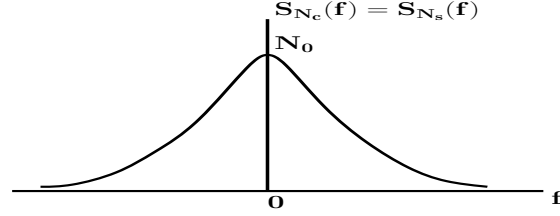


Fig. 3: Spectral density of in-phase and quadrature components of the output of RLC filter.

Q. 2. Ans. a). Modulation index of the FM wave is equal to 10.

b). The capture range is equal to 3.16 KHz.

c). The FM signal $s(t)$ is given by

$$s(t) = A_c \cos(2\pi \times 10^7 t + \sin(10^3 \pi t)) .$$

d). Use Carson's rule to compute bandwidth. Approximate bandwidth of the FM signal is equal to 11 KHz.

e). The peak amplitude of carrier is equal to 14.14 volt.

f). The FM signal in terms of Bessel function is given by

$$s(t) = 14.14 \sum_{-\infty}^{\infty} J_n(10) \cos(2\pi(f_c + n f_m)t) .$$

g). The peak amplitude of carrier is equal to 2.5 volt.

h). The positive side of magnitude spectrum of the WBFM signal is shown in Figure 4. In it, the center frequency component is f_c and other frequency components are $f_c + n f_m$, where $n = 1, 2, \dots$

Q. 3. Ans. a). The input noise power density is shown in Figure 5.

b). Assuming unit gain, the ideal bandpass filter frequency response is plotted in Figure 6.

c). The bandpass filtered output noise spectral density is shown Figure 7.

d). Mean square value (MSV) of the output noise process is equal to the average power which is nothing but area under spectral density.

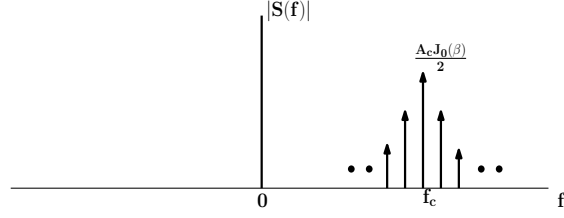


Fig. 4: Magnitude spectrum of the FM signal.

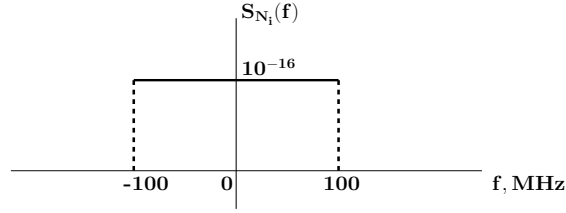


Fig. 5: Input noise spectral density.

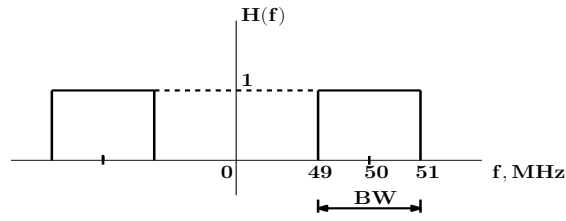


Fig. 6: Frequency response of an ideal BPF.

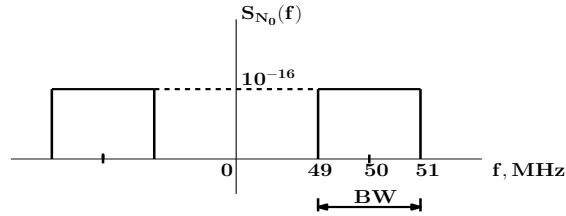


Fig. 7: Bandpass filtered output noise spectral density.

Thus, the MSV is equal to 0.4 nW or 93.98 dB or 63.98 dBm.

e). By symmetry of $S_{N_0}(f)$, spectral densities of in-phase and quadrature components of output noise are equal as shown Figure 8 below.

f). The impulse response of the filter is given by

$$h(t) = 4B \operatorname{sinc}(2Bt) \cos(2\pi f_c t),$$

where $B = 2$ MHz, and $f_c = 50$ MHz.

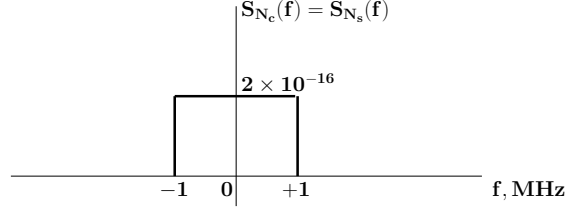


Fig. 8: Spectral density of in-phase and quadrature components of the output of ideal BPF.

Q. 4. Ans. a). $a = 2b$.

b). Probability is equal to $\frac{1}{2} \left(\exp\left(-\frac{1}{b}\right) - \exp\left(-\frac{2}{b}\right) \right)$.

c). The mean value of Y is equal to zero. The mean square value of Y is equal to $2b^2$ or $\frac{a^2}{2}$.

d). CDF of Y is given by

$$F_Y(y) = \begin{cases} \frac{1}{2} \exp\left(\frac{y}{b}\right), & y < 0, \\ 1 - \frac{1}{2} \exp\left(-\frac{y}{b}\right), & y \geq 0 \end{cases}.$$

CDF of Y is given by $F^c_Y(y) = 1 - F_Y(y)$.

e). The moment generating function is given by

$$M_Y(t) = \frac{1}{1 - b^2 t^2}, |t| < \frac{1}{b}.$$